Chapter 9. Cluster Analysis: Basic Concepts and Methods

- Cluster Analysis: Basic Concepts
- Partitioning Methods
- Hierarchical Methods
- Density-Based Methods 
- Expectation Maximization
- Evaluation of Clustering
- Summary

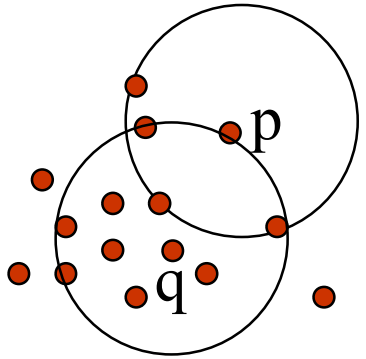
Density-Based Clustering Methods

- Clustering based on density (local cluster criterion), such as density-connected points
- Major features:
  - Discover clusters of arbitrary shape
  - Handle noise
  - One scan
  - Need density parameters as termination condition
- Several interesting studies:
  - DBSCAN: Ester, et al. (KDD'96)
  - OPTICS: Ankerst, et al (SIGMOD'99).
  - DENCLUE: Hinneburg & D. Keim (KDD'98)
  - CLIQUE: Agrawal, et al. (SIGMOD'98) (more grid-based)

Density-Based Clustering: Basic Concepts

- Two parameters:
  - *Eps*: Maximum radius of the neighbourhood
  - *MinPts*: Minimum number of points in an Eps-neighbourhood of that point
- $N_{Eps}(p)$: {q belongs to D | dist(p,q) ≤ Eps}
- Directly density-reachable: A point *p* is directly density-reachable from a point *q* w.r.t. *Eps*, *MinPts* if
  - *p* belongs to $N_{Eps}(q)$
  - core point condition:

$$|N_{Eps}(q)| \geq MinPts$$

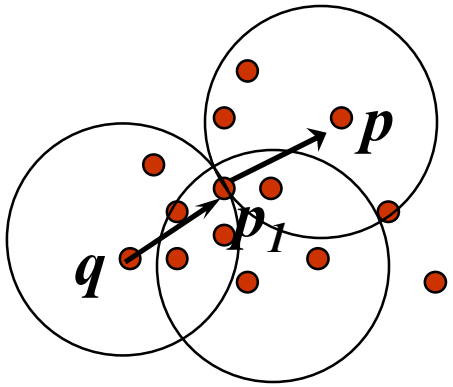


MinPts = 5

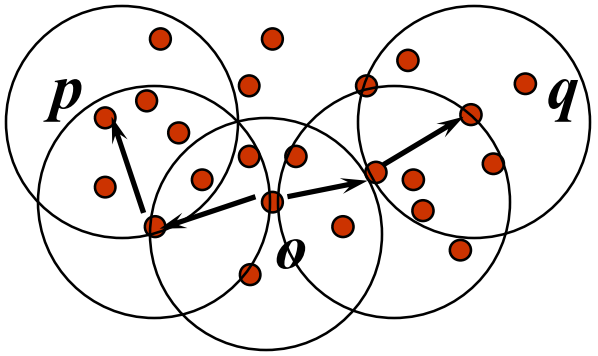
Eps = 1 cm

Density-Reachable and Density-Connected

- Density-reachable:
  - A point *p* is density-reachable from a point *q* w.r.t. *Eps*, *MinPts* if there is a chain of points p_1, …, p_n, p_1 = *q*, p_n = *p* such that p_{i+1} is directly density-reachable from p_i

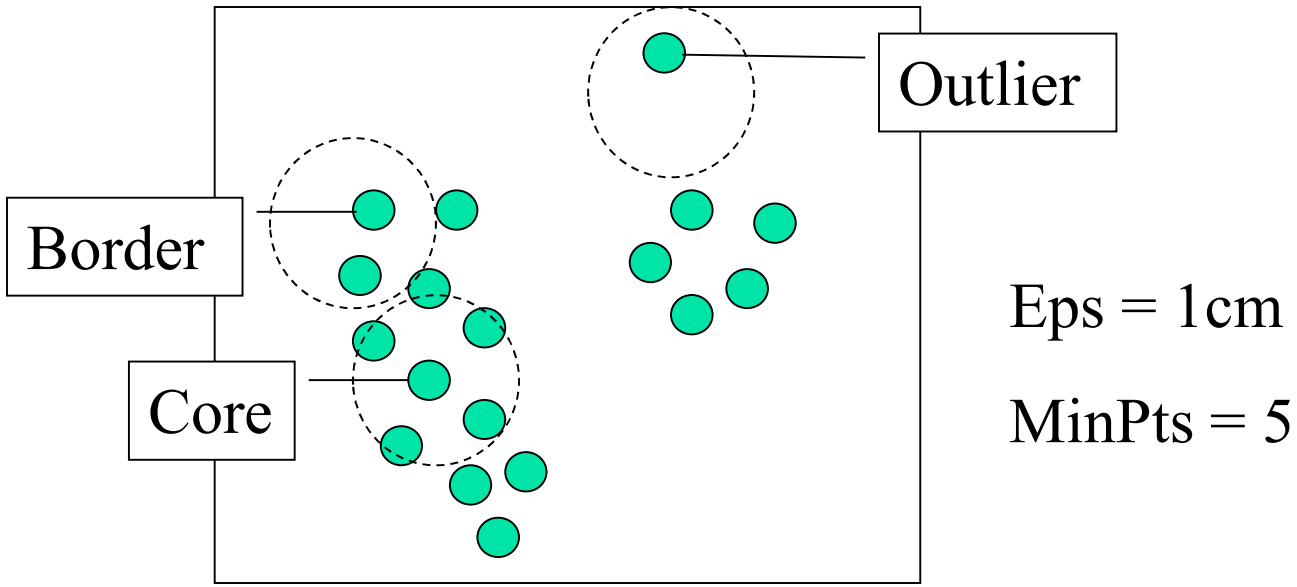


- Density-connected
  - A point *p* is density-connected to a point *q* w.r.t. *Eps*, *MinPts* if there is a point *o* such that both, *p* and *q* are density-reachable from *o* w.r.t. *Eps* and *MinPts*



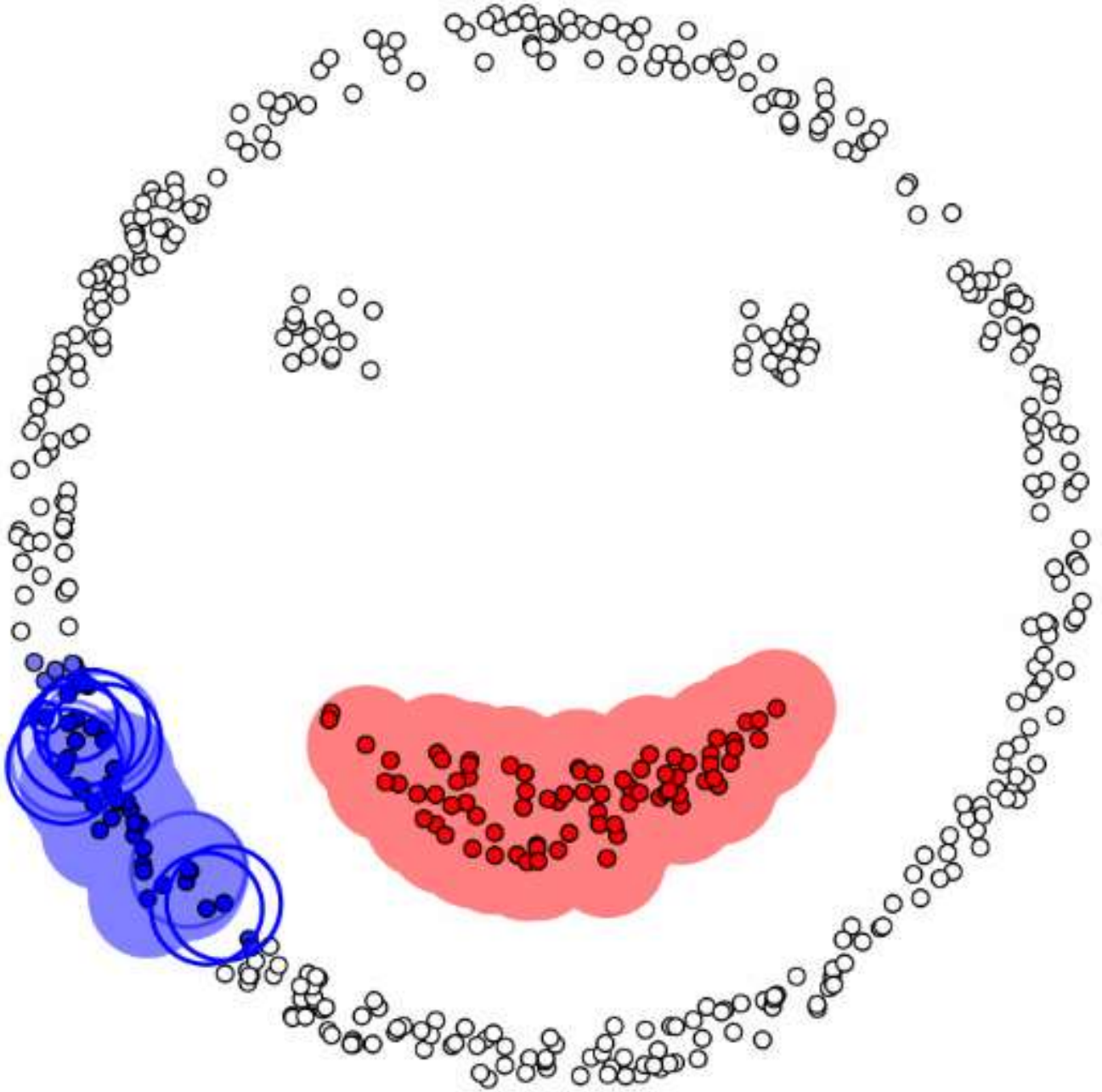
DBSCAN: Density-Based Spatial Clustering of Applications with Noise

- Relies on a *density-based* notion of cluster: A *cluster* is defined as a maximal set of density-connected points
- Discovers clusters of arbitrary shape in spatial databases with noise



DBSCAN: The Algorithm

- Arbitrary select a point *p*
- Retrieve all points density-reachable from *p* w.r.t. *Eps* and *MinPts*
- If *p* is a core point, a cluster is formed
- If *p* is a border point, no points are density-reachable from *p* and DBSCAN visits the next point of the database
- Continue the process until all of the points have been processed



DBSCAN: Sensitive to Parameters

Figure 8. DBScan results for DS1 with MinPts at 4 and Eps at (a) 0.5 and (b) 0.4.

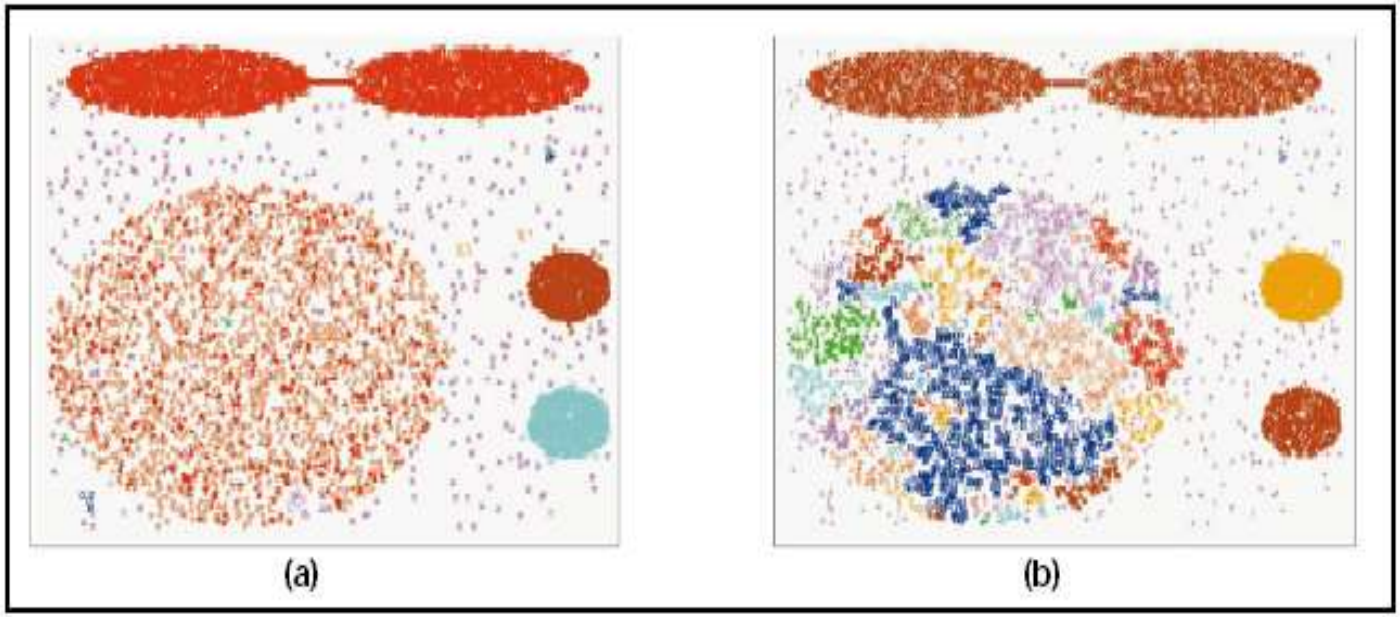
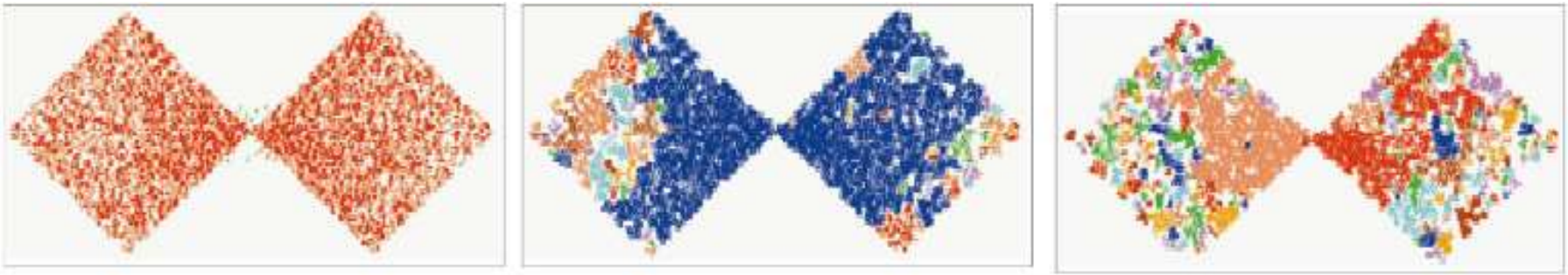


Figure 9. DBScan results for DS2 with MinPts at 4 and Eps at (a) 5.0, (b) 3.5, and (c) 3.0.



Extensions to DBSCAN

- OPTICS: Ordering Points To Identify the Clustering Structure
  - Ankerst, Breunig, Kriegel, and Sander (SIGMOD'99)
- DENCLUE
  - DENsity-based CLUstEring by Hinneburg & Keim (KDD'98)

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Model-based Approach

- A model is hypothesized for each of the clusters and tries to find the best fit of that model to each other

- Example: EM / GM Clustering
  - EM: Expectation-Maximization
  - Gaussian Mixtures

Mixture Models in 1-D

- Case 1
  - Assume the data points come from two Gaussians with unknown parameters
  - We know each point comes from which Gaussian

$$\mu_b = \frac{x_1 + x_2 + \ldots + x_{n_b}}{n_b}$$

$$\sigma_b^2 = \frac{(x_1 - \mu_b)^2 + \ldots + (x_n - \mu_b)^2}{n_b}$$



Figure © V. Lavrenko, Lecture on Clustering, University of of Edinburg

Mixture Models of 1-D

- Case 2
  - We don't know the sources
  - We know the parameters of the Gaussians



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Mixture Models of 1-D

- Case 2
  - We don't know the sources
  - We know the parameters of the Gaussians

$$P(x_i \mid b) = \frac{1}{\sqrt{2\pi\sigma_b^2}} \exp\left\{ -\frac{(x_i - \mu_i)^2}{2\sigma_b^2} \right\}$$

$$P(b \mid x_i) = \frac{P(x_i \mid b)P(b)}{P(x_i \mid b)P(b) + P(x_i \mid a)P(a)}$$

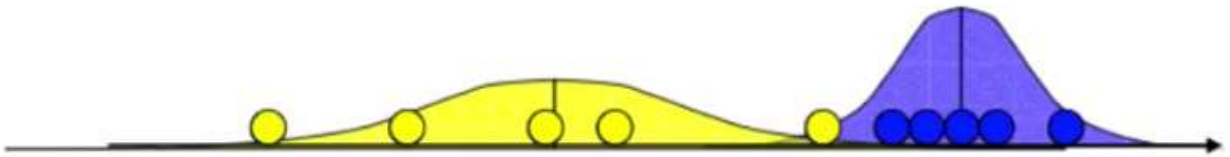


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Expectation-Maximization

- Chicken and egg problem!
  - We need parameters to guess source of points
  - We need to know source to estimate parameters
- EM algorithm
  - Start with two randomly placed Gaussians
  - REPEAT
    - Use the Gaussians to determine which point comes from which Gaussian (Case 2)
    - Adjust parameters to fit points assigned to them (Case 1)
  - UNTIL Convergence

Expectation-Maximization (cont.)

- EM algorithm
  - Start with two randomly placed Gaussians
  - REPEAT
    - Use the Gaussians to determine which point comes from which Gaussian (E-step)
    - Adjust parameters to fit points assigned to them (M-step)
  - UNTIL Convergence

EM Visualization

- Initialization
  - Initialize all parameters randomly
  - Uniform distribution for priors

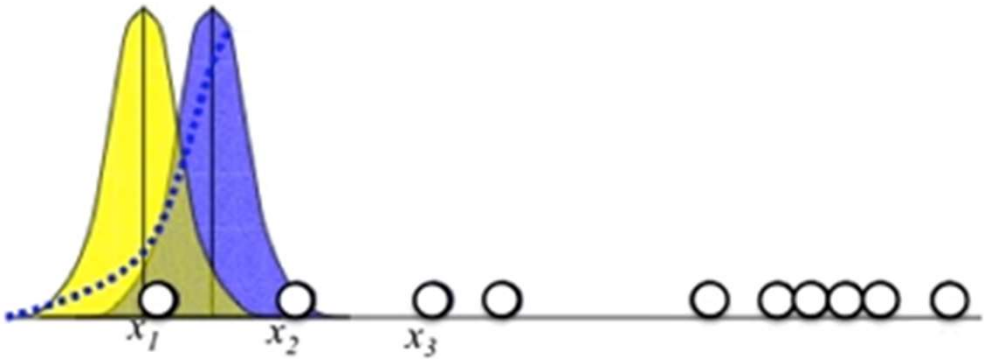


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EM Visualization (cont.)

$$P(x_i \mid b) = \frac{1}{\sqrt{2\pi\sigma_b^2}} \exp\left\{-\frac{(x_i - \mu_b)^2}{2\sigma_b^2}\right\}$$

$$b_i = P(b \mid x_i) = \frac{P(x_i \mid b)P(b)}{P(x_i \mid b)P(b) + P(x_i \mid a)P(a)}$$

$$a_i = P(a \mid x_i) = 1 - b_i$$

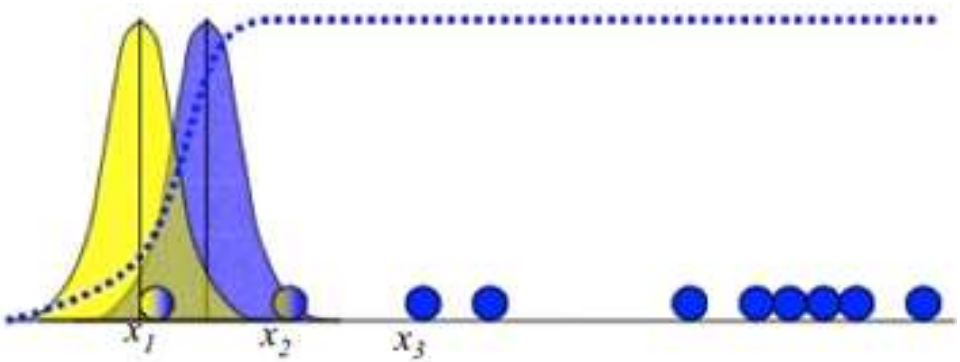
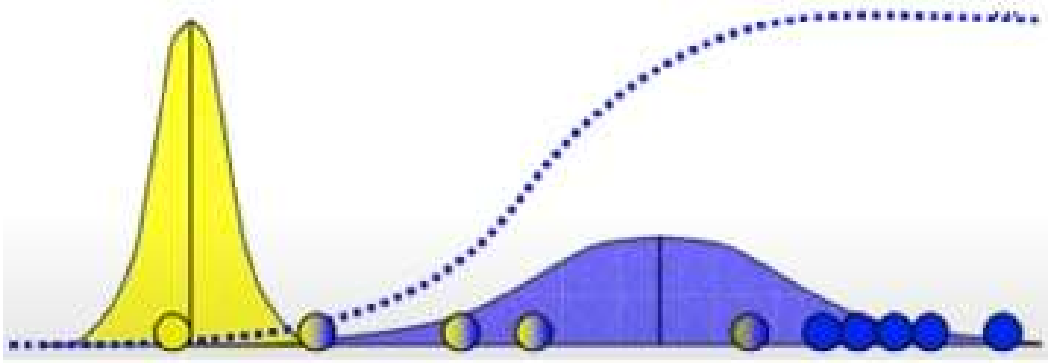


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EM Visualization (cont.)

$$\mu_b = \frac{b_1 x_1 + b_2 x_2 + \ldots + b_n x_n}{b_1 + b_2 + \ldots + b_n}$$

$$\sigma_b^2 = \frac{b_1 (x_1 - \mu_b)^2 + \ldots + b_n (x_n - \mu_b)^2}{b_1 + b_2 + \ldots + b_n}$$



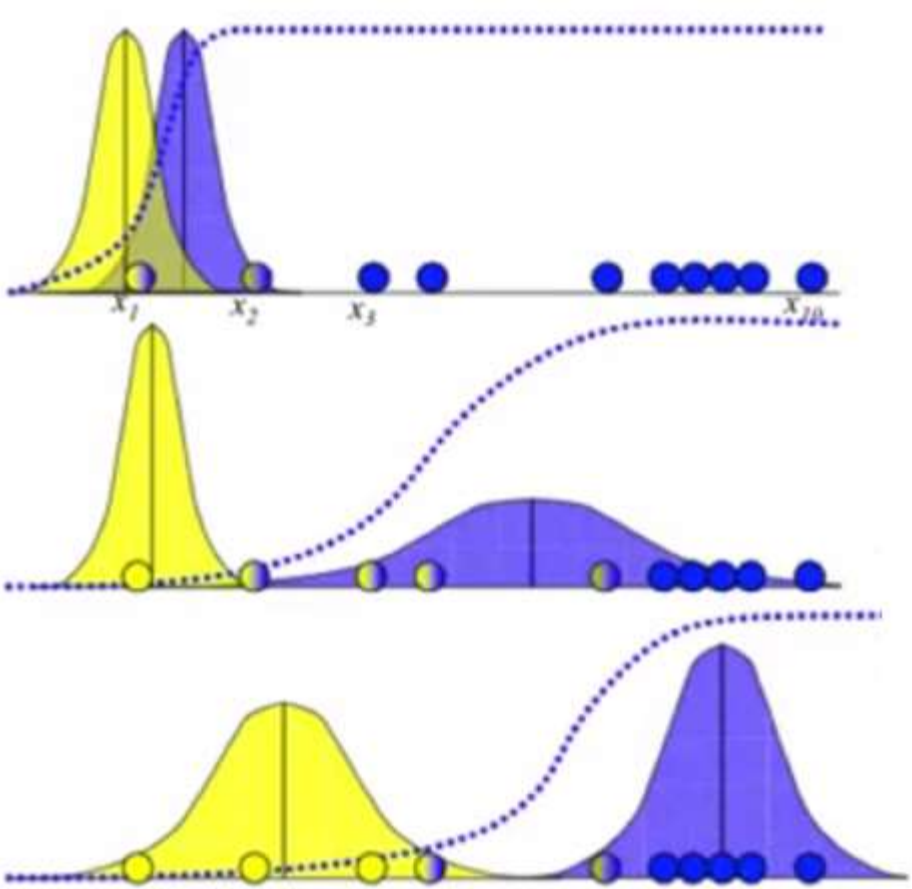
could also estimate priors:

$P(b) = (b_1 + b_2 + \ldots b_n) / n$

$P(a) = 1 - P(b)$

EM Visualization (cont.)

- The process continues until convergence



References (1)

- R. Agrawal, J. Gehrke, D. Gunopulos, and P. Raghavan. Automatic subspace clustering of high dimensional data for data mining applications. SIGMOD'98
- M. Ankerst, M. Breunig, H.-P. Kriegel, and J. Sander. Optics: Ordering points to identify the clustering structure, SIGMOD'99.
- Beil F., Ester M., Xu X.: "Frequent Term-Based Text Clustering", KDD'02
- M. M. Breunig, H.-P. Kriegel, R. Ng, J. Sander. LOF: Identifying Density-Based Local Outliers. SIGMOD 2000.
- M. Ester, H.-P. Kriegel, J. Sander, and X. Xu. A density-based algorithm for discovering clusters in large spatial databases. KDD'96.
- M. Ester, H.-P. Kriegel, and X. Xu. Knowledge discovery in large spatial databases: Focusing techniques for efficient class identification. SSD'95.
- D. Gibson, J. Kleinberg, and P. Raghavan. Clustering categorical data: An approach based on dynamic systems. VLDB'98.
- V. Ganti, J. Gehrke, R. Ramakrishan. CACTUS Clustering Categorical Data Using Summaries. KDD'99.

References (2)

- D. Gibson, J. Kleinberg, and P. Raghavan. Clustering categorical data: An approach based on dynamic systems. In Proc. VLDB'98.
- S. Guha, R. Rastogi, and K. Shim. Cure: An efficient clustering algorithm for large databases. SIGMOD'98.
- S. Guha, R. Rastogi, and K. Shim. ROCK: A robust clustering algorithm for categorical attributes. In *ICDE'99*, pp. 512-521, Sydney, Australia, March 1999.
- A. Hinneburg, D.l A. Keim: An Efficient Approach to Clustering in Large Multimedia Databases with Noise. KDD'98.
- A. K. Jain and R. C. Dubes. Algorithms for Clustering Data. Printice Hall, 1988.
- G. Karypis, E.-H. Han, and V. Kumar. CHAMELEON: A Hierarchical Clustering Algorithm Using Dynamic Modeling. *COMPUTER*, 32(8): 68-75, 1999.
- T. Zhang, R. Ramakrishnan, and M. Livny. BIRCH : An efficient data clustering method for very large databases. SIGMOD'96
- E. Knorr and R. Ng. Algorithms for mining distance-based outliers in large datasets. VLDB'98.